

Chapter 93

Blur Nodes

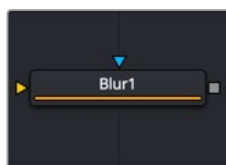
This chapter details the Blur nodes available in Fusion. The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Blur [Blur]



The Blur node

Blur Node Introduction

The Blur node does exactly what its name implies – it blurs the input image. This is one of the most commonly used image-processing operations.

Inputs

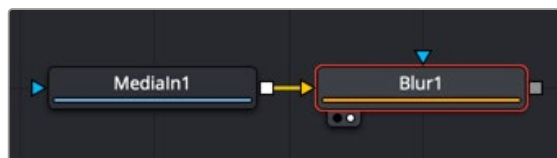
The two inputs on the Blur node are used to connect a 2D image and an effect mask that can be used to limit the blurred area.

Input: The orange input is used for the primary 2D image that is blurred.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the blur to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

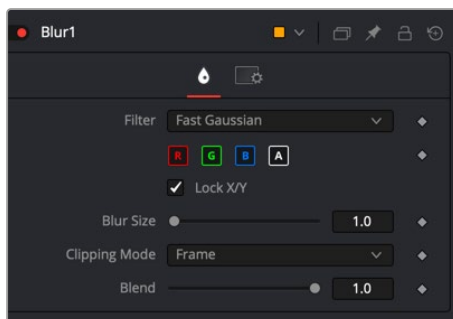
Basic Node Setup

The Blur node, like many 2D image-processing nodes, receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Blur node applied to a MediaIn1 node

Inspector



Blur controls

NOTE: Since a perfect Gaussian filter would require examining an infinite number of pixels, all practical Gaussians are, of necessity, approximations. The algorithm Fusion uses is a highly-optimized approach that has many strengths, but can create visible ringing around the edges in certain extreme cases. This ringing appears only when blurring float-depth images and is normally far below the limits of visibility, especially in final renders or HiQ mode, but may appear in subsequent processing. If you experience this, selecting the Multi-box filter may be a good choice.

Controls Tab

The Controls tab contains the primary controls necessary for customizing the blur operation, including five filter algorithms.

Filter

The Filter menu is where you select the type of filter used to create the blur.

- **Box Blur:** This option is faster than the Gaussian blur but produces a lower-quality result.
- **Bartlett:** This option is a more subtle, anti-aliased blur filter.
- **Multi-box:** Multi-box uses a Box filter layered in multiple passes to approximate a Gaussian shape. With a moderate number of passes (e.g., four), a high-quality blur can be obtained, often faster than the Gaussian filter and without any ringing.
- **Gaussian:** Gaussian applies a smooth, symmetrical blur filter, using a sophisticated constant-time Gaussian approximation algorithm.
- **Fast Gaussian:** Gaussian applies a smooth, symmetrical blur filter, using a sophisticated constant-time Gaussian approximation algorithm. This mode is the default filter method.

Color Channels (RGBA)

The filter defaults to operating on R, G, B, and A channels. Selective channel filtering is possible by clicking each channel button to make them active or inactive.

NOTE: This is not the same as the RGBA checkboxes found under the common controls. The node takes these selections into account before it processes the image, so deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab are applied after the node has processed.

Lock X/Y

Locks the X and Y Blur sliders together for symmetrical blurring. This is enabled by default.

Blur Size

Sets the amount of blur applied to the image. When the Lock X and Y control is deselected, independent control over each axis is provided.

Clipping Mode

This option determines how edges are handled when performing domain-of-definition rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

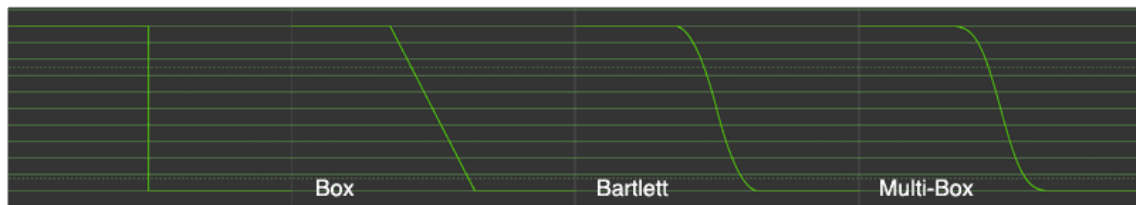
Blend

The Blend slider determines the percentage of the affected image that is mixed with original image. It blends in more of the original image as the value gets closer to 0.

This control is a cloned instance of the Blend slider in the Common Controls tab. Changes made to this control are simultaneously made to the one in the common controls.

Examples

Following is a comparison of Blur filters visualized as “cross-sections” of a filtered edge. As you can see, Box creates a linear ramp, while Bartlett creates a somewhat smoother ramp. Multi-box and Gaussian are indistinguishable unless you zoom in really close on the slopes. They both lead to even smoother ramps, but as mentioned above, Gaussian overshoots slightly and may lead to negative values if used on floating-point images.



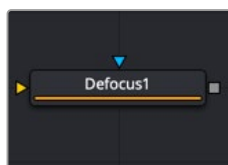
Blur filters visualized as “cross sections” of a filtered edge

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Defocus [Dfo]



The Defocus node

Defocus Node Introduction

The Defocus node simulates the effects of an out-of-focus camera lens, including blooming and image flaring. It provides a fast Gaussian mode, as well as a more realistic but slower Lens mode.

Inputs

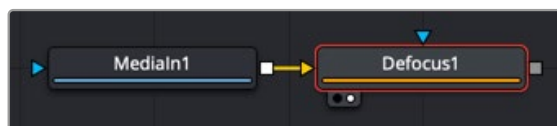
The two inputs on the Defocus node are for connecting a 2D image and an effect mask that can be used to limit the simulated defocused area.

Input: The orange input is used for the primary 2D image for defocusing.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the defocus to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

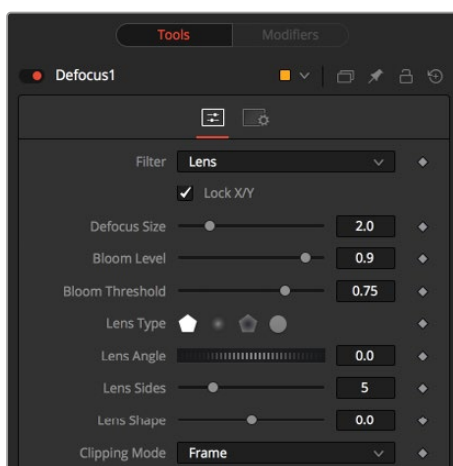
Basic Node Setup

The Defocus node receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Defocus node applied to a MediaIn1 node

Inspector



Defocus controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the defocus operation.

Filter

Use this menu to select the exact method applied to create the defocus. Gaussian applies a simplistic effect, while Lens mode creates a more realistic defocus. Lens mode takes significantly longer than Gaussian.

Lock X/Y

When Lock X/Y is selected, this performs the same amount of defocusing to both the X- and Y-axis of the image. Deselect to obtain individual control.

Defocus Size

The Defocus Size control sets the size of the defocus effect. Higher values blur the image by greater amounts and produce larger blooms.

Bloom Level

The Bloom Level control determines the intensity and size of the blooming applied to pixels that are above the bloom threshold.

Bloom Threshold

Pixels with values above the set Bloom Threshold are defocused and have a glow applied (blooming). Pixels below that value are only defocused.

The following four lens options are available only when the Filter is set to Lens.

- **Lens Type:** The basic shape used to create the “bad bokeh” effect. This can be refined further with the Angle, Sides, and Shape sliders.
- **Lens Angle:** Defines the rotation of the shape. Best visible with NGon lens types. Because of the round nature of a circle, this slider has no visible effect when the Lens Type is set to Circle.
- **Lens Sides:** Defines how many sides the NGon shapes have. Best visible with NGon lens types. Because of the round nature of a circle, this slider has no visible effect when the Lens Type is set to Circle.
- **Lens Shape:** Defines how pointed the NGons are. Higher values create a more pointed, starry look. Lower values create smoother NGons. Best visible with NGon lens types and Lens Sides between 5 and 10. Because of the round nature of a circle, this slider has no visible effect when the Lens Type is set to Circle.

Clipping Mode

This option determines how edges are handled when performing domain-of-definition rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node’s domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node’s effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node’s effect that would normally be outside the upstream DoD is treated as black/transparent.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Directional Blur [DrBl]



The Directional Blur node

Directional Blur Node Introduction

This node is used to create Directional and Radial blurs. It is useful for creating simulated motion blur and light ray-type effects. Directional Blur affects all channels (RGBA).

Inputs

The two inputs on the Directional Blur node are used to connect a 2D image and an effect mask which can be used to limit the blurred area.

Input: The orange input is used for the primary 2D image that has the directional blur applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the directional blur to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

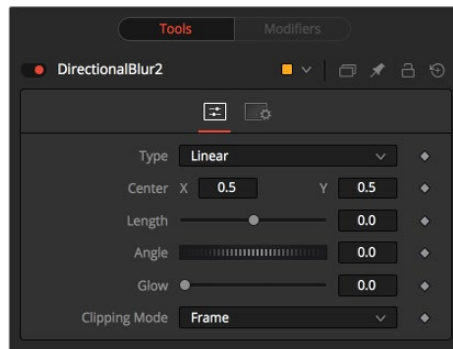
Basic Node Setup

The Directional Blur node receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Directional Blur node applied to a MediaIn1 node

Inspector



Directional Blur controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the directional blur operation.

Type

This menu is used to select the type of directional blur to be applied to the image.

- **Linear:** Linear distorts the image in a straight line, resembling the scenery that appears in the window of a speeding train.
- **Radial:** Radial creates a distortion that originates at some arbitrary center, radiating outward the way that a view would appear if one were at the head of the train looking forward.
- **Centered:** The Centered button produces a similar result to linear, but the blur effect is equally distributed on both sides of the original.
- **Zoom:** Zoom creates a distortion in the scale of the image smear to simulate the zoom streaking of a camera filming with a slow shutter speed.

Center X and Y

This coordinate control and related viewer crosshair affects the Radial and Zoom Motion Blur types only. It is used to position where the blurring effect starts.

Length

Length adjusts the strength and heading of the effect. Values lower than zero cause blurs to head opposite the angle control. Values greater than the slider maximum may be typed into the slider's edit box.

Angle

In both Linear and Center modes, this control modifies the direction of the directional blur. In the Radial and Zoom modes, the effect is similar to the camera spinning while looking at the same spot. If the setting of the length slider is other than zero, the effect creates a whirlpool effect.

Glow

This adds a Glow to the directional blur, which can be used to duplicate the effect of increased camera exposure to light caused by longer shutter speeds.

Clipping Mode

This option determines how edges are handled when performing domain-of-definition rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Glow [Glo]



The Glow nozzle

Glow Node Introduction

A Glow is created by blurring an image, and then brightening the blurred result and mixing it back with the original. The Glow node provides a variety of variations on this theme. For example, a Bartlett glow is a high-quality glow with a smoother drop-off; however, it is more processor-intensive at larger sizes.

Inputs

The Glow node has three inputs: an orange one for the primary 2D image input, a blue one for an effect mask, and a third white input for a Glow mask.

Input: The orange input is used for the primary 2D image that has the glow applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the source of the glow to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

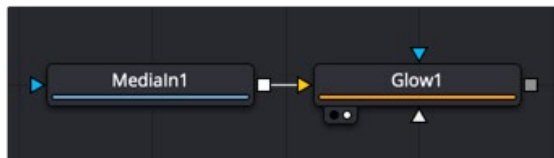
Glow Mask: The Glow node supports pre-masking using the white glow mask input. A Glow pre-mask filters the image before applying the glow. The glow is then merged back over the original image. This is different from a regular effect mask that clips the rendered result.

The Glow mask allows the glow to extend beyond the borders of the mask, while restricting the source of the glow to only those pixels within the mask.

Glow masks are identical to Effect masks in every other respect.

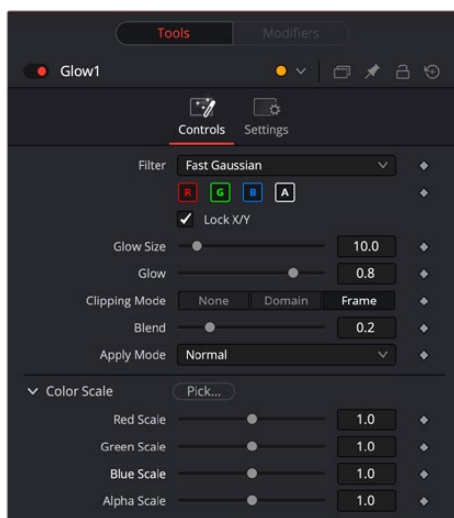
Basic Node Setup

The Glow node receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Glow node applied to a MediaIn1 node in DaVinci Resolve

Inspector



Glow controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the glow operation. A Color Scale section at the bottom of the Inspector can be used for tinting the glow.

Filter

Use this menu to select the method of Blur used in the filter. The selections are described below.

- **Box:** A simple but very fast Box filter.
- **Bartlett:** Bartlett adds a softer, subtler glow with a smoother drop-off but may take longer to render than Box.
- **Multi-box:** Multi-box uses a Box filter layered in multiple passes to approximate a Gaussian shape. With a moderate number of passes (e.g., four), a high-quality blur can be obtained, often faster than the Gaussian filter, and without any ringing.
- **Gaussian:** Gaussian adds a soft glow, blurred by the Gaussian algorithm.
- **Fast Gaussian:** Fast Gaussian adds a soft glow, blurred by the Gaussian algorithm. This is the default method.
- **Blend:** Blend adds a nonlinear glow that is evenly visible in the whites and blacks.
- **Hilight:** Hilight adds a glow without creating a halo in the surrounding pixels.
- **Solarize:** Solarize adds a glow and solarizes the image.

Color Channels (RGBA)

This filter defaults to operating on R, G, B, and A channels. Selective channel filtering is possible by clicking each channel to make them active or inactive.

NOTE: This is not the same as the RGBA checkboxes found under the common controls. The node takes these selections into account before it processes the image, so deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab are applied after the node has processed.

Lock X/Y

When Lock X/Y is checked, both the horizontal and vertical glow amounts are locked. Otherwise, separate amounts of glow may be applied to each axis.

Glow Size

Glow Size determines the size of the glow effect. Larger values expand the size of the glowing highlights of the image.

Num Passes

Only available in Multi-box mode. Larger values lead to a smoother distribution of the effect, but also increase render times. It's good to find the line between desired quality and acceptable render times.

Glow

The Glow slider determines the intensity of the glow effect. Larger values tend to completely blow the image out to white.

Clipping Mode

This option determines how edges are handled when performing domain-of-definition rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Blend

The Blend slider determines the percentage of the affected image that is mixed with original image. It blends in more of the original image as the value gets closer to 0.

This control is a cloned instance of the Blend slider in the Common Controls tab. Changes made to this control are simultaneously made to the one in the common controls.

Apply Mode

Three Apply Modes are available when it comes to applying the glow to the image.

- **Normal:** Default. This mode simply adds the glow directly over top of the original image.
- **Merge Under:** Merge Under places the glow beneath the image, based on the Alpha channel. Threshold mode permits clipping of the threshold values.
- **Threshold:** This control clips the effect of the glow. A new range slider appears. Pixels in the glowed areas with values below the low value are pushed to black. Pixels with values greater than high are pushed to white.
- **High-Low Range Control:** Available only in Threshold mode. Pixels in the glowed areas with values below the low value are pushed to black. Pixels with values greater than high are pushed to white.

Color Scale (RGBA)

These Scale sliders can be used to adjust the amount of glow applied to each color channel individually, by tinting the glow.

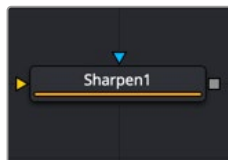
By click and holding on the Pick button, then dragging the pointer over the viewer, you can select a specific color from the image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Sharpen [Shrp]



The Sharpen node

Sharpen Node Introduction

The Sharpen node uses a convolution filter to enhance detail in an image overall or to an individual channel.

Inputs

The two inputs on the Sharpen node are used to connect a 2D image and an effect mask that can limit the area affected by the sharpen.

Input: The orange input is used for the primary 2D image for sharpening.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the sharpen to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

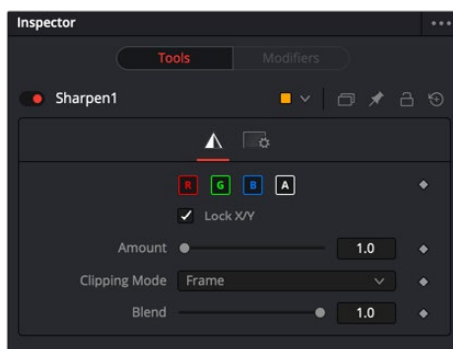
Basic Node Setup

The Sharpen node receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Sharpen node applied to a MediaIn1 node in DaVinci Resolve

Inspector



Sharpen controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the sharpen operation.

Color Channels (RGBA)

This filter defaults to operating on R, G, B, and A channels. Selective channel filtering is possible by clicking the channel buttons to make them active or inactive.

NOTE: This is not the same as the RGBA checkboxes found under the common controls. The node takes these selections into account before it processes the image, so deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab are applied after the node has processed.

Lock X/Y

This locks the X and Y Sharpen sliders together for symmetrical sharpening. This is checked by default.

Amount

This slider sets the amount of sharpening applied to the image. When the Lock X/Y control is deselected, independent control over each axis is provided.

Clipping Mode

This option determines how edges are handled when performing domain-of-definition rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Blend

The Blend slider determines the percentage of the affected image that is mixed with original image. It blends in more of the original image as the value gets closer to 0.

This control is a cloned instance of the Blend slider in the Common Controls tab. Changes made to this control are simultaneously made to the one in the common controls.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Soft Glow [SGlo]



The Soft Glow node

Soft Glow Node Introduction

The Soft Glow node is similar to the Glow node but performs additional processing of the image to create a much softer, more natural glow.

This node is perfect for atmospheric haze around planets, skin tones, and simulating dream like environments.

Inputs

Like the Glow node, Soft Glow also has three inputs: an orange one for the primary image input, a blue one for an effect mask, and a third white input for a Glow mask.

Input: The orange input is used for the primary 2D image for the soft glow.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the soft glow to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

Glow Mask: The Soft Glow node supports pre-masking using the white glow mask input. A Glow pre-mask filters the image before applying the soft glow. The soft glow is then merged back over the original image. This is different from a regular effect mask that clips the rendered result.

The Glow mask allows the soft glow to extend beyond the borders of the mask, while restricting the source of the soft glow to only those pixels within the mask.

Glow masks are identical to effect masks in every other respect.

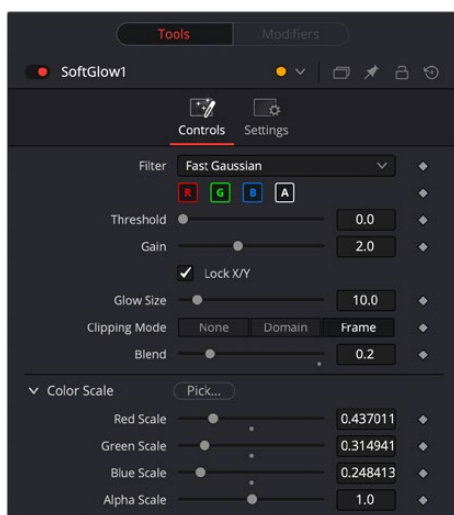
Basic Node Setup

The Soft Glow node receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Soft Glow node applied to a MediaIn1 node

Inspector



Soft Glow controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the soft glow operation. A color scale section at the bottom of the Inspector can be used for tinting the soft glow.

Filter

Use this menu to select the method of Blur used in the filter. The selections are described below.

- **Box:** A simple but very fast Box filter.
- **Bartlett:** Bartlett adds a softer, subtler glow with a smoother drop-off but may take longer to render than Box.
- **Multi-box:** Multi-box uses a Box filter layered in multiple passes to approximate a Gaussian shape. With a moderate number of passes (e.g., four), a high-quality blur can be obtained, often faster than the Gaussian filter and without any ringing.
- **Gaussian:** Gaussian adds a soft glow, blurred by the Gaussian algorithm. This is the default method.

Color Channels (RGBA)

The filter defaults to operating on R, G, B, and A channels. Selective channel filtering is possible by clicking the channel buttons to make them active or inactive.

NOTE: This is not the same as the RGBA checkboxes found under the common controls. The node takes these selections into account before it processes the image, so deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab are applied after the node has processed.

Threshold

This control is used to limit the effect of the soft glow. The higher the threshold, the brighter the pixel must be before it is affected by the glow.

Gain

The Gain control defines the brightness of the glow.

Lock X/Y

When Lock X/Y is checked, both the horizontal and vertical glow amounts are locked. Otherwise, separate amounts of glow may be applied to each axis of the image.

Glow Size

This amount determines the size of the glow effect. Larger values expand the size of the glowing highlights of the image.

Num Passes

Available only in Multi-box mode. Larger values lead to a smoother distribution of the effect, but also increase render times. It's good to find the line between desired quality and acceptable render times.

Clipping Mode

This option determines how edges are handled when performing domain-of-definition rendering. This is profoundly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Blend

The Blend slider determines the percentage of the affected image that is mixed with original image. It blends in more of the original image as the value gets closer to 0.

This control is a cloned instance of the Blend slider in the Common Controls tab. Changes made to this control are simultaneously made to the one in the common controls.

Color Scale (RGBA)

These Scale sliders are used to adjust the amount of glow applied to each color channel individually, by tinting the glow.

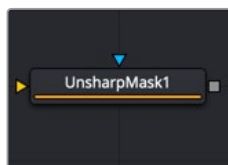
By click and holding on the Pick button, then dragging the pointer over the viewer, you can select a specific color from the image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Unsharp Mask [UsM]



The Unsharp Mask node

Unsharp Mask Introduction

Unsharp masking is a technique used to sharpen only the edges within an image. This node is most often used to correct for blurring and loss of detail in low-contrast images; for example, to extract useful detail from long exposure shots of faraway galaxies.

This filter extracts a range of frequencies from the image and blurs them to reduce detail. The blurred result is then compared to the original images. Pixels with a significant difference between the original and the blurred image are likely to be an edge detail. The pixel is then brightened to enhance it.

Inputs

The two inputs on the Unsharp Mask node are used to connect a 2D image and an effect mask for limiting the effect.

Input: The orange input is used for the primary 2D image for the Unsharp Mask.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Unsharp Mask to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

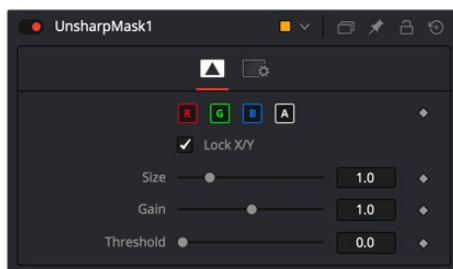
Basic Node Setup

The Unsharp Mask node receives a 2D image like the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



An Unsharp mask node applied to a MediaIn1 node

Inspector



Unsharp Mask controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the Unsharp Mask operation.

Color Channels (RGBA)

The filter defaults to operating on R, G, B, and A channels. Selective channel filtering is possible by clicking the channel buttons to make them active or inactive.

NOTE: This is not the same as the RGBA checkboxes found under the common controls. The node takes these selections into account before it processes the image, so deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab are applied after the node has processed.

Lock X/Y

When Lock X/Y is checked, both the horizontal and vertical sharpen amounts are locked. Otherwise, separate amounts of glow may be applied to each axis of the image.

Size

This control adjusts the size of blur filter applied to the extracted image. The higher this value, the more likely it is that pixels are identified as detail.

Gain

The Gain control adjusts how much gain is applied to pixels identified as detail by the mask. Higher values create a sharper image.

Threshold

This control determines the frequencies from the source image to be extracted. Raising the value eliminates lower-contrast areas from having the effect applied.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Vari Blur [VBI]



The Vari Blur node

Vari Blur Node Introduction

The Vari Blur node gives a true per-pixel variable blur, using a second image to control the amount of blur for each pixel. It is somewhat similar in effect to the Depth Blur node but uses a different approach for frequently cleaner results.

Inputs

There are two inputs on the Vari Blur node for the primary image: the blur map image, and an effect mask.

Input: The gold image input is a required connection for the primary image you wish to blur.

Blur Image: The green input is also required, but it can accept a spline shape, text object, still image, or movie file as the blur map image. Once connected, you can choose red, green, blue, Alpha, or luminance channel to create the shape of the blur.

Effect Mask: The optional blue effect mask input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Vari Blur to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

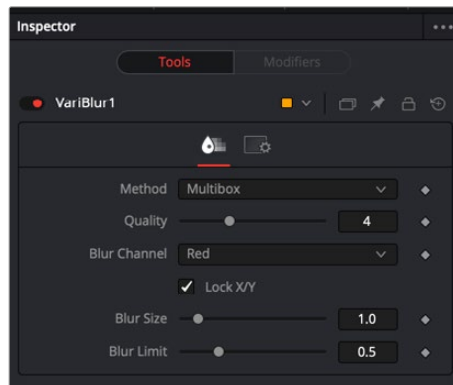
Basic Node Setup

The Vari Blur node receives a 2D image like the MediaIn1 shown below. A gradient Background tool connects to the Blur image input to control the areas affected by the blur. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Vari Blur node applied to a MediaIn1 node and a gradient background directing the blurred regions

Inspector



Vari Blur controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the Vari Blur operation.

Method

Use this menu to select the method of Blur used in the filter. The selections are described below.

- **Soften:** This method varies from a simple Box shape to a Bartlett triangle to a decent-looking Smooth blur as Quality is increased. It is a little better at preserving detail in less-blurred areas than Multi-box.
- **Multi-box:** Similar to Soften, this gives a better Gaussian approximation at higher Quality settings.
- **Defocus:** Produces a flat, circular shape to blurred pixels that can approximate the look of a defocus.

Quality

Increasing Quality gives smoother blurs, at the expense of speed. Quality set to 1 uses a very fast but simple Box blur for all Method settings. A Quality of 2 is usually sufficient for low Blur Size values. A Quality of 4 is generally good enough for most jobs unless Blur Size is particularly high.

Blur Channel

This selects which channel of the Blur Image controls the amount of blurring applied to each pixel.

Lock X/Y

When selected, only a Blur Size control is shown, and changes to the amount of blur are applied to both axes equally. If the checkbox is cleared, individual controls appear for both X and Y Blur Size.

Blur Size

Increasing this control increases the overall amount of blur applied to each pixel. Those pixels where the Blur image is black or nonexistent are blurred, despite the Blur Size.

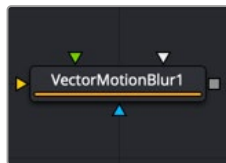
Blur Limit

This slider limits the useable range from the Blur image. Some Z-depth images can have values that go to infinity, which skew blur size. The Blur Limit is a way to keep values within range.

Common Controls**Settings Tab**

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Vector Motion Blur [VMB]



The Vector Motion Blur node

Vector Motion Blur Introduction

This node is used to create directional blurs based on a Motion Vector map or AOV (Arbitrary Output Variable) channels exported from 3D-rendering software like Arnold, Renderman, or V-Ray. You can also generate motion vectors using the Optical Flow node in Fusion.

The vector map is typically two floating-point images: one channel specifies how far the pixel is moving in X, and the other specifies how far the pixel is moving in Y. These channels may be embedded in OpenEXR or RLA/RPF images, or may be provided as separate images using the node's Vectors input.

The vector channels should use a float16 or float32 color depth, to provide + and – values.

A value of 1 in the X channel would indicate that pixel has moved one pixel to the right, while a value of –10 indicates ten pixels of movement to the left.

Inputs

The Vector Motion Blur node has three inputs for a 2D image, a motion vector pass, and an effect mask.

Input: The required orange input is for a 2D image that receives the motion blur.

Vectors: The green input is also required. This is where you connect a motion vector AOV rendered from a 3D application or an EXR file generated from the Optical Flow node in Fusion.

Vector Mask: The white Vector Mask input is an optional input that masks the image before processing.

Effect Mask: The common blue input is used for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the source of the motion blur to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

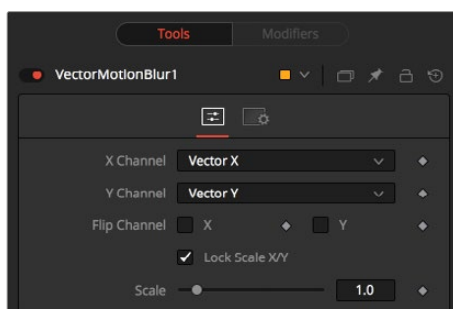
Basic Node Setup

The Vector Motion Blur node receives a 2D image like the IMAGE shown below. A MediaIn or Loader node containing motion vectors is connected to the Vector's input. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Vector Motion Blur node applied to a MediaIn or Loader node with motion vectors connected to the Vectors input

Inspector



Vector Motion Blur node

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the Vector Motion Blur operation.

X Channel

Use this menu to select which channel of the image provides the vectors for the movement of the pixels along the X-axis.

Y Channel

Use this menu to select which channel of the image provides the vectors for the movement of the pixels along the Y-axis.

Flip Channel

These checkboxes can be used to flip, or invert, the X and Y vectors. For instance, a value of 5 for a pixel in the X-vector channel would become -5 when the X checkbox is enabled.

Lock Scale X/Y

Selecting this checkbox provides access to separate sliders for X and Y Scale. By default, only a single Scale slider is provided.

Scale

The X and Y vector channel values for a pixel are multiplied by the value of this slider. For example, given a scale of 2 and a vector value of 10, the result would be 20. This slider splits to show Scale X and Scale Y if the Lock Scale X/Y checkbox is not enabled.

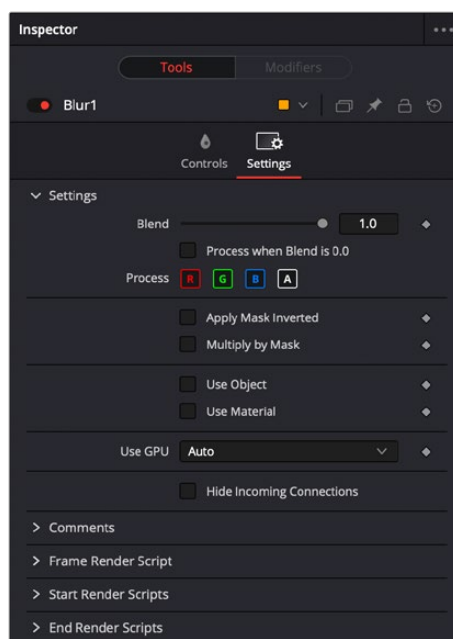
Common Controls**Settings Tab**

The Settings tab in the Inspector is also duplicated in other Blur nodes. These common controls are described in the following “The Common Controls” section.

The Common Controls

Nodes that handle blur operations share several identical controls in the Inspector. This section describes controls that are common among Blur nodes.

Inspector



Common Blur settings

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Blur category. The Settings controls are even found on third-party Blur-type plugin tools. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools where deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not in the mask (i.e. set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports object and material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific material ID or object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Use GPU

The GPU menu has three settings. Disable turns off GPU hardware accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A Quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Comments

The Comments field is used to add notes to a tool. Click in the field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.